

AICHU TAN

Dublin, CA | 408-650-2150 | dtan3697@sdsu.edu | [aichutan.github.io](https://github.com/aichutan) | linkedin.com/in/aichutan

PROFESSIONAL SUMMARY

Data Scientist with experience building and deploying end-to-end machine learning and AI systems across public health, agriculture, and urban analytics. Specialized in time-series forecasting, computer vision (YOLO), and LLM-powered applications (LangChain, RAG). Proven ability to develop production-ready solutions using Streamlit, Hugging Face, and AWS, with strong expertise in feature engineering, model evaluation, and translating complex data into actionable insights.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript, C++

Machine Learning: XGBoost, Random Forest, Scikit-learn, Clustering, K-Prototypes

Deep Learning & Time Series: LSTM, ARIMA, Exponential Smoothing, TensorFlow, YOLOv8

Data & Visualization: Pandas, NumPy, Tableau, Matplotlib, ArcGIS

Cloud & MLOps: AWS (EC2, S3, RDS), MLflow, Git, Hugging Face, Streamlit, Supabase, Render

PROFESSIONAL EXPERIENCE

Data Science Researcher - SDSU DiMoLab

Sep 2025 - Present

- Developed **time-series forecasting models using Python** to predict outbreak trends, improving early-risk detection and enabling data-driven public health planning.
- Performed **data cleaning, feature engineering, and outbreak risk modeling** on epidemiological and climate datasets to improve predictive performance.
- Contributed to the development of an **AI-based early warning system for disease surveillance**.
- Co-authoring a research manuscript for submission to a peer-reviewed journal.

AI Researcher - Metabolism of Cities Living Lab

Feb 2025 - Dec 2025

- Designed and deployed an end-to-end AI pipeline from data processing and model training to real-time inference and user-facing applications.
- Built a computer vision system using YOLOv8 trained on 13.5K+ images (20 classes), achieving **0.894 mAP@50**.
- Developed an LLM-powered retrieval system (RAG, LangChain) to generate actionable agricultural recommendations.
- Launched a production-ready web application using Streamlit and Hugging Face Spaces for real-time usage.
- Co-authored research accepted at the **4th International One Health Conference (Rome, 2025)**.

Business Owner - Wholesale Clothing

Apr 2009 - Oct 2016

- Increased sales by 50% through data-driven market analysis and customer segmentation.
- Optimized supply chain operations using forecasting and analytics.
- Managed international import/export operations across Taiwan, China, and Indonesia.

Process Engineer - ProMOS Technology

Jul 2004 - Feb 2009

- Improved DRAM yield via **statistical process control (SPC)** and root cause analysis.
- Reduced manufacturing defects through data modeling and process optimization.

SELECTED PROJECTS

End-to-End Dengue Forecasting System | Python, LSTM, XGBoost, Random Forest, Streamlit

- Built a time-series forecasting system using 15+ years of epidemiological and climate data, achieving **R² up to 0.89**.
- Engineered lag-based and temporal features to improve model accuracy.
- Identified key drivers (rainfall, humidity, lagged incidence) to support proactive decision-making.
- Deployed an interactive Streamlit dashboard for real-time monitoring.

AI for Farmers: Plant Disease Detection & Advisory System | YOLOv8, LangChain, Streamlit, Hugging Face

- Developed a full-stack AI system combining computer vision (YOLOv8) and LLM-based retrieval (RAG).
- Achieved **0.894 mAP@50** on 13,500+ labeled images across 20 classes.
- Built a LangChain pipeline to translate agricultural research into actionable insights.
- Deployed a real-time application using Streamlit and Hugging Face Spaces.

Predicting Restaurant Ratings Using Yelp Metadata | MLflow, XGBoost, RF, Pandas

- Built a machine learning pipeline to classify restaurant success (≥ 4 ★ vs < 4 ★) using 36,000+ Yelp business records.
- Engineered 61 features and applied **5-fold cross-validation**.
- Tracked experiments using **MLflow** to compare model performance.
- Optimized **XGBoost**, achieving **0.733 accuracy** and **0.79 ROC-AUC**.
- Conducted feature importance and clustering to uncover consumer behavior patterns and restaurant archetypes.

Mapping for Change - Urban Inequality Analysis | R, Tableau, Spatial Analysis, Time Series, Data Visualization

- Built an interactive ArcGIS dashboard to analyze **homelessness, housing affordability, and environmental inequality** across Downtown San Diego.
- Integrated 5+ public datasets (ACS, 311, PITC, Tree Equity Score) and applied spatial analysis (heatmaps, choropleths) to identify inequality clusters.
- Uncovered key relationships between **poverty, rent burden, and reduced tree equity**, highlighting areas of environmental injustice.
- Delivered a public-facing web application to support **data-driven urban planning and policy decisions**.

EDUCATION

M.S. in Big Data Analytics, San Diego State University - **GPA 4.0**

Aug 2024 - Dec 2025

Google Data Analytics Professional Certificate

Dec 2022

B.S. in Chemical Engineering, National Cheng Kung University (Taiwan)

Sept 1999 - June 2003